

Registration No: -

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Total Number of Pages: 02

Course: B.Tech
Sub Code: REC3C002

3rd Semester Regular / Back Examination: 2022-23

SUB: SIGNALS AND SYSTEMS

BRANCH(S): ECE, ETC

MAX MARKS: 100

TIME: 3 HOURS

Q CODE: L516

Answer Question No.1 (Part-1) which is compulsory, any eight from Part-II and any two from Part-III.

The figures in the right hand margin indicate marks.

Part- I

Q1 Answer All-10

(02x10)

- Find Z-transform of $u(n-1)$.
- Differentiate between autocorrelation and cross correlation.
- What is the superposition theorem to check linearity of a discrete time system?
- How to check causality and stability of a discrete time system?
- Write the formulation for continuous time Fourier transform.
- What is the relation between DFT and z-transform?
- What is the long division method in inverse Z-Transform?
- Determine the DFT of $2\delta(n)+3$.
- Write any two properties of z-transform.
- Differentiate between linear and circular convolution.

Part- II

Q2 Answer Any Eight out of Twelve

(06x08)

- Determine the energy and power of the signal $y[n] = 10 \cos\left(10n + \frac{\pi}{8}\right)$.
- Find the linear convolution of $x(n) = \{1, 4, 16, 32\}$ and $h(n) = \{1, 2, 3, 4\}$ without using the tabular method.
- Using the properties of Z transform, find the Z transform of the following
 - $x[n] = u[-n]$
 - $x[n] = 2^n u[n-2]$
- Determine the Z-transform of $x(n) = \{-3, -1, -8, 1, 2, 7, 3, 9, 15, 7\}$
- Find the cross correlation of $x(n) = \{1, -1, 1, -1\}$ and $h(n) = \{2, 2, 3, 3\}$
- Find the inverse Z-Transform of $H(z) = \frac{2z}{9z^2 - 16z + 1}$, $|z| > 1$ using long division method.
- Show that energy of power signal is ∞ .

- h) A causal system has input $\delta[n]$ and output $y[n]$. Find the frequency response and impulse response of the causal system if, $y[n] = \delta[n] + \frac{1}{6}\delta[n-1] - \frac{1}{6}\delta[n-2]$
- i) Write any four properties of DFT.
- j) How to find inverse z-transform using power series expansion method?
- k) Find the circular convolution of $x(n) = \{1, 2, 3\}$ & $h(n) = \{1, 1, 1\}$ using DFT method.
- l) Compute the DFT of $x[n] = \cos \frac{n\pi}{2}$ where $N = 4$.

Part-III

Answer Any Two out of Four

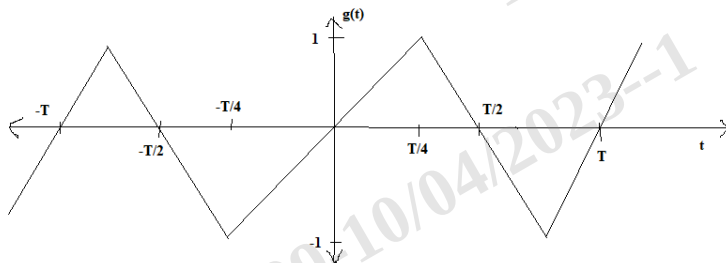
Q3 Determine the inverse z-transform of $X(z) = \frac{z}{-4z + 3z^2 + 1}$, if the ROC (16)

(A) $|z| > 1$

(B) $|z| < \frac{1}{3}$

(C) $1 > |z| > \frac{1}{3}$

Q4 Find the trigonometric Fourier series of the triangular waveform shown below in the following figure. (16)



Q5 Write notes of (any two) (16)

1. Properties of Continuous Time Fourier Transform
2. Discrete Time Systems Described by Difference Equations
3. Inverse Z-Transform Methods

Q6 Check the linearity, causality and time-invariance of the systems represented by (16)

(i) $g[n] = 12y[n] + \frac{30y[n]}{y[n-4]}$

(ii) $g[n] = F\{y[n]\} = 11y^2[n-5] + ny[n-1]$